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Saveetha Institute of Medical And Technical Sciences
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AI-Driven Citizen Support and Complaint Management Using Natural Language Processing

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfillment for the award of the degree of

DSA0306-Natural Language Processing

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted by

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Under the Supervision of

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DECLARATION

We, **A. Mohith Krishna(192425199)** of the Department of Computer Science Engineering, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled **AI-Driven Citizen Support and Complaint Management Using Natural Language Processing** is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place: Chennai

Date:

Signature of the Students with Names

A. Mohith Krishna(192425199)

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled **AI-Driven Citizen Support and Complaint Management Using Natural Language Processing** has been carried out by **A. Mohith Krishna(192425199)** under the supervision of **Dr. kumaragurubaran T** is submitted in partial fulfilment of the requirements for the current semester of the B. Tech **Artificial Intelligence and Machine Learning** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

The AI-Driven Citizen Support and Complaint Management Using Natural Language Processing project is developed to provide an intelligent, automated, and efficient approach for handling citizen complaints. In traditional complaint management systems, complaints are often reviewed and processed manually, which can lead to delays, incorrect department assignment, difficulty in identifying urgent issues, repeated complaints, and challenges in processing complaints submitted in different languages. The proposed system addresses these problems by integrating Natural Language Processing and machine learning techniques into a single complaint management workflow. When a citizen submits a complaint, the system first identifies the language using FastText and translates supported multilingual complaints into a common processing language using NLLB. The translated complaint is then analyzed using BERT to understand its context and classify it into the appropriate department, such as water, electricity, roads, sanitation, drainage, or waste management. Random Forest is used to determine the priority level of the complaint as Low, Medium, High, or Critical based on factors such as urgency, service disruption, safety risk, and overall impact rather than sentiment. The system further uses SBERT to generate meaningful sentence representations and Cosine Similarity to compare new complaints with previously stored complaints, helping identify semantically similar or duplicate complaints. BART is used to generate concise summaries containing the important information from lengthy complaints so that officials can understand the issue quickly. After the complaint is processed, it can be routed to the relevant department for further action and monitoring. The system can also maintain complaint information such as department, priority, status, and previous complaint records, supporting better organization and decision-making. By combining language detection, translation, classification, priority detection, semantic similarity, summarization, and complaint routing, the proposed system provides an end-to-end solution for intelligent citizen support. Overall, the project demonstrates the practical application of NLP and machine learning in a real-world public service environment and aims to reduce manual effort, improve complaint accuracy, identify repeated issues, support multilingual citizens, prioritize important complaints, and help government departments respond to citizen problems in a faster, more organized, and effective manner.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description
AI	Artificial Intelligence
NLP	Natural Language Processing
BERT	Bidirectional Encoder Representations from Transformers
NLLB	No Language Left Behind
RF	Random Forest
SBERT	Sentence-Bidirectional Encoder Representations from Transformers
BART	Bidirectional and Auto-Regressive Transformers
ML	Machine Learning

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Citizen complaint management is an important part of public administration because people regularly report issues related to water supply, damaged roads, electricity, sanitation, drainage, waste management, streetlights, transportation, and other civic services. Traditional complaint systems generally depend on citizens manually choosing a complaint category and administrators manually checking and forwarding complaints to the appropriate department.

When a large number of complaints are received, manual processing becomes time-consuming and may result in incorrect classification, delayed forwarding, duplicate complaints, and difficulty in identifying urgent issues. Citizens may also submit complaints in different regional languages, creating additional challenges for complaint processing.

Natural Language Processing provides methods for automatically understanding written complaints. The proposed AI-Driven Citizen Support and Complaint Management System uses NLP models to identify the complaint language, translate multilingual complaints, determine the responsible department, identify complaint priority, detect duplicate complaints, and generate concise summaries. The system therefore provides an intelligent and automated approach to citizen complaint handling.

1.2 Need for the Project

Existing complaint management systems require considerable manual effort for reading, categorizing, prioritizing, and forwarding complaints. When thousands of complaints are received, this process can become inefficient.

The major problems include incorrect department selection by citizens, delays in identifying critical complaints, repeated submission of similar complaints, difficulties in understanding

complaints submitted in regional languages, and lengthy complaint descriptions that take additional time for officials to read.

The proposed system addresses these problems through automated NLP-based complaint processing. It can help citizens submit complaints more easily and assist government departments in processing complaints faster and more consistently.

The project is significant because it combines multiple NLP techniques into a single complaint-management workflow rather than solving only one individual task.

1.3 Problem Statement

Existing citizen complaint management systems often depend on manual classification and manual forwarding of complaints. Citizens may select the wrong department, submit complaints in different languages, submit repeated complaints for the same problem, or describe problems using lengthy and unstructured text.

Therefore, there is a need for an intelligent complaint-management system capable of automatically understanding complaint text, identifying its language, translating it when necessary, assigning the appropriate department, determining the level of priority based on the seriousness of the issue, detecting semantically duplicate complaints, and generating concise summaries.

The engineering challenge is to integrate multiple NLP models while maintaining acceptable classification accuracy, processing speed, scalability, reliability, and usability.

1.4 Project Aim

The main aim of this project is to develop an AI-driven citizen support and complaint management system using Natural Language Processing that automatically processes citizen complaints and assists authorities in efficient complaint classification, prioritization, duplicate detection, summarization, and department routing.

1.5 Project Objectives

1. To automatically identify the language of a citizen complaint using FastText Language Identification.
2. To translate complaints written in regional languages into English using the NLLB multilingual translation model.
3. To classify complaints into appropriate government departments using a fine-tuned BERT model.
4. To classify complaints into Low, Medium, High, and Critical priority levels using the Random Forest algorithm.
5. To detect similar or duplicate complaints using Sentence-BERT embeddings and Cosine Similarity.
6. To automatically generate short complaint summaries using the BART summarization model.
7. To route the processed complaint to the appropriate department and provide complaint-tracking facilities.
8. To evaluate the developed system using suitable NLP classification and performance metrics.

1.6 Scope

The project focuses mainly on text-based citizen complaints.

The system includes complaint submission, multilingual language identification, translation, department classification, priority detection, duplicate complaint detection, automatic summarization, department routing, complaint storage, status tracking, and administration facilities.

The project considers departments such as:

- Water Supply
- Roads
- Electricity

- Sanitation
- Drainage
- Waste Management
- Street Lights
- Public Transport
- Health
- Other Civic Services

The current project primarily deals with textual complaints. Image-based complaint analysis, voice complaint recognition, physical infrastructure monitoring, and direct control of government infrastructure are outside the current scope.

1.7 Expected Engineering Outcome

The expected engineering outcome is a functional software prototype capable of accepting citizen complaints and automatically processing them through an NLP pipeline.

The expected workflow is:

Citizen Complaint → Language Detection → Translation → Department Classification → Priority Classification → Duplicate Detection → Summarization → Department Routing → Complaint Tracking

The final system is expected to reduce manual complaint-processing effort and provide faster, more organized complaint management.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review focuses on research articles, NLP models, complaint-management systems, text-classification methods, multilingual NLP models, semantic similarity techniques, and text-summarization approaches.

Relevant work can be identified through research databases such as Google Scholar, IEEE Xplore, Springer, ScienceDirect, and official model documentation. Particular attention is given to Transformer-based NLP models and machine-learning approaches used for classification and semantic understanding.

2.2 Review of Existing Work

Traditional Complaint Management Systems

Traditional citizen grievance systems allow users to submit complaints through websites or mobile applications. However, many systems depend on users to manually choose complaint categories and departments. Incorrect category selection can result in delayed processing.

Keyword-Based Text Classification

Earlier complaint-classification systems often used keywords or rule-based approaches. For example, words such as "water", "road", or "electricity" could be used for categorization.

These methods are simple and computationally inexpensive. However, they cannot effectively understand the complete meaning of sentences and may fail when similar words are used in different contexts.

Machine-Learning Classification

Algorithms such as Naive Bayes, Support Vector Machines, Decision Trees, and Random Forest have been used for text classification. These approaches provide improved performance when compared with simple rule-based systems but often require manually generated features.

Transformer-Based Classification

Transformer models such as BERT provide contextual representations of text. BERT can understand the relationship between words in a sentence and can therefore classify complaints more accurately than keyword-based methods.

For example:

“There are large potholes on my road.”

can be understood as a Roads Department complaint.

Similarly:

“There is no drinking water supply in our street.”

can be classified under the Water Department.

Multilingual NLP

Citizen complaints may be submitted in several regional languages. Language-identification models such as FastText can automatically identify the input language.

Multilingual translation models such as **NLLB – No Language Left Behind** allow complaints written in supported regional languages to be translated into a common processing language.

Duplicate Complaint Detection

Traditional duplicate detection based only on exact word matching cannot identify complaints that express the same meaning using different words.

Sentence-BERT generates semantic vector representations of complete sentences. These embeddings can then be compared through Cosine Similarity.

For example:

Complaint 1: “No water supply in Gandhi Street.”

Complaint 2: “Water has not been available in Gandhi Street.”

Although the sentences are different, they describe a similar problem.

Automatic Complaint Summarization

Officials may receive lengthy complaint descriptions. Transformer-based generative models such as BART can convert lengthy text into a short summary while preserving the main issue.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed System
Manual complaint classification	Simple to implement	Slow and dependent on human effort	Replaced with automated classification
Keyword-based classification	Fast and inexpensive	Poor contextual understanding	BERT provides better context
Traditional machine learning	Good for structured features	Requires feature engineering	Random Forest is suitable for priority prediction
Exact-text duplicate checking	Easy to implement	Cannot understand semantic similarity	SBERT improves duplicate detection
Manual translation	Accurate when performed by experts	Slow and expensive	NLLB provides automatic translation
Manual summarization	Can produce good summaries	Time-consuming	BART automates summarization

2.4 Research / Engineering Gap

Many existing complaint-management systems primarily provide facilities for complaint registration and status tracking. They do not combine multiple NLP capabilities into one integrated solution.

Important gaps include:

- Limited multilingual complaint handling.
- Dependence on manual complaint categorization.
- Limited automatic priority identification.
- Weak semantic duplicate detection.
- Lack of automatic complaint summarization.
- Lack of an integrated NLP pipeline.

2.5 Proposed Contribution

The proposed project integrates several NLP and machine-learning techniques into a single citizen complaint-management system.

The major contribution is the combined processing pipeline:

FastText → NLLB → BERT → Random Forest → SBERT → Cosine Similarity → BART →
Department Routing

This architecture enables the system to automatically understand, classify, prioritize, compare, summarize, and route citizen complaints.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder Requirements

The main stakeholders are citizens, system administrators, government departments, and complaint-management authorities.

Citizens require an easy method for submitting complaints and checking complaint status.

Government departments require correctly categorized complaints, priority information, concise summaries, and tools for updating complaint status.

Administrators require facilities for monitoring complaints, users, departments, system statistics, and model performance.

Functional Requirements

Requirement	Type	Target
Citizen complaint submission	Functional	Complaint successfully registered
Automatic language identification	Functional	Identify supported input language
Complaint translation	Functional	Convert supported language into English
Department classification	Functional	Predict appropriate department
Priority classification	Functional	Low/Medium/High/Critical
Duplicate detection	Functional	Identify semantically similar complaints
Complaint summarization	Functional	Generate short meaningful summary
Complaint tracking	Functional	Display latest complaint status

Non-Functional Requirements

Requirement	Type	Target
Accuracy	Non-Functional	High classification accuracy
Response time	Non-Functional	Reasonable processing time
Security	Non-Functional	Authorized access to administrative functions
Reliability	Non-Functional	Stable complaint storage and processing
Usability	Non-Functional	Simple user interface
Scalability	Non-Functional	Ability to process increasing complaint volume

3.2 Constraints & Applicable Standards

The important project constraints include computational resources, availability of labelled complaint datasets, model inference time, multilingual translation quality, privacy of citizen information, and storage requirements.

The system must also follow responsible AI practices by avoiding unnecessary collection of personal information and restricting administrative access.

Where applicable, standard software-development practices, secure authentication methods, data-protection principles, and recognized software documentation practices should be followed.

3.3 Design Specifications

Parameter	Target
Supported complaint input	Text
Language identification	FastText
Translation	NLLB
Department classification	BERT
Priority levels	Low, Medium, High, Critical
Priority algorithm	Random Forest
Duplicate detection	SBERT + Cosine Similarity
Summarization	BART
Database	MySQL/PostgreSQL
User interface	Web-based
Access	Citizen, Department, Administrator

3.4 Alternative Solutions & Evaluation

Alternative 1 – Rule-Based System

The system can classify complaints using predefined keywords and rules. It is simple and fast but performs poorly when complaints contain ambiguous or context-dependent language.

Alternative 2 – Traditional Machine-Learning System

Algorithms such as Naive Bayes, SVM, and Random Forest can classify complaints. These approaches provide reasonable performance but usually require feature engineering.

Alternative 3 – Transformer-Based NLP System

Transformer models provide contextual understanding and can effectively perform classification, translation, semantic comparison, and summarization. Their primary disadvantages are larger computational and memory requirements.

The proposed system therefore combines Transformer models with traditional machine learning where each is most suitable.

3.5 Selected Design & Detailed Design

The selected system follows a modular architecture.

Step 1 – Complaint Submission

A citizen enters complaint details through the web interface.

Step 2 – Language Detection

FastText analyzes the input and identifies the complaint language.

Step 3 – Translation

If the complaint is not already in English, NLLB translates it into English.

Step 4 – Department Classification

BERT analyzes the complaint meaning and predicts the responsible department.

Step 5 – Priority Classification

Random Forest evaluates severity-related information and classifies the complaint as Low, Medium, High, or Critical.

Step 6 – Duplicate Detection

SBERT converts complaints into sentence embeddings. Cosine Similarity measures semantic similarity between the new complaint and existing complaints.

Step 7 – Summarization

BART generates a short summary containing the main complaint information.

Step 8 – Routing

The complaint is stored and forwarded to the predicted department.

System Flow

Citizen → Complaint Submission → FastText → NLLB → BERT → Random Forest → SBERT
+ Cosine Similarity → BART → Database → Department Dashboard

3.6 Trade-off Analysis

Design Decision	Alternative	Benefit	Disadvantage	Final Decision
BERT classification	Keyword rules	Better contextual understanding	Higher computation	BERT selected
Random Forest priority	Rules	Handles multiple severity features	Requires labelled data	Random Forest selected
SBERT similarity	Exact matching	Semantic comparison	Additional computation	SBERT selected
NLLB translation	Manual translation	Automatic multilingual support	Model size	NLLB selected
BART summarization	Extract first sentence	Better summaries	Computational requirement	BART selected

3.7 Sustainability, Ethics, Safety, Risk & Security

Sustainability

The proposed digital system reduces dependence on paper-based complaint management and can automate repetitive administrative activities.

Ethics

Citizen information should be collected only when required. AI predictions should assist authorities rather than blindly replace responsible administrative decision-making.

Health and Safety

Complaints involving dangerous infrastructure, electrical hazards, severe water contamination, accidents, or similar risks should receive appropriate priority.

Security

The system should provide secure login, role-based access, protected passwords, database security, session management, and restricted administrative functions.

Risk Register

Risk		Likelihood	Severity	Mitigation
Incorrect department classification		Medium	Medium	Train and evaluate BERT with quality data
Incorrect priority		Medium	High	Use labelled severity data and human review
Translation error		Medium	Medium	Preserve original complaint with translation
Duplicate false detection		Medium	Medium	Tune cosine-similarity threshold
Unauthorized system access		Low	High	Authentication and role-based authorization
Model failure		Low	Medium	Provide fallback/manual processing

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project follows a modular software-development approach. Individual NLP components are first developed and evaluated separately. The components are then connected through a common backend to create the complete complaint-processing pipeline.

The main resources include complaint datasets, multilingual text datasets, Python NLP libraries, pretrained Transformer models, a database, and a web-development framework.

4.2 Software Implementation & Modern Tools Used

Frontend

The citizen and administrator interfaces can be developed using React, HTML, CSS, JavaScript/TypeScript, and Tailwind CSS.

Backend

Python-based frameworks such as Flask or FastAPI can be used to expose the NLP functions through REST APIs.

NLP Implementation

FastText

Used to identify the language of the complaint.

NLLB

Used to translate supported regional languages into English.

BERT

Fine-tuned using labelled citizen complaints to classify complaints into departments.

Random Forest

Trained using complaint severity information to predict Low, Medium, High, or Critical priority.

SBERT

Generates semantic embeddings of complaint sentences.

Cosine Similarity

Compares the new complaint embedding with previous complaint embeddings.

BART

Generates a concise summary of lengthy complaints.

Database

MySQL or PostgreSQL can store:

- Citizen details
- Complaint information
- Department details
- Priority
- Complaint status
- Complaint summaries
- Complaint history
- Duplicate information

Tools

Tool	Purpose
Python	NLP and backend programming
Hugging Face Transformers	Pretrained NLP models
Scikit-learn	Random Forest and evaluation
FastText	Language identification
React	Frontend development
MySQL/PostgreSQL	Database
VS Code	Development environment
Git/GitHub	Version control

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance
Language identification	Submit multilingual complaints	Correct language identified
Translation	Compare source and translated meaning	Main meaning retained
Department classification	Test labelled complaint samples	Correct department predicted
Priority prediction	Test severity-labelled complaints	Correct priority predicted
Duplicate detection	Test similar/different complaints	Similar complaints detected
Summarization	Test long complaints	Meaningful concise summary generated
Database storage	Submit complaint	Complaint stored correctly

4.4 Results

The result section should contain the actual values obtained after implementation.

Recommended results to include are:

- BERT classification accuracy.
- Precision.
- Recall.
- F1-score.
- Random Forest priority accuracy.
- Confusion matrix.
- Language identification accuracy.
- Translation examples.
- Duplicate similarity scores.
- BART-generated summaries.
- Average complaint-processing time.

Example Result Format

Module	Algorithm	Expected Output
Language Detection	FastText	Detected language
Translation	NLLB	English complaint
Department Classification	BERT	Department name
Priority Classification	Random Forest	Priority level
Duplicate Detection	SBERT + Cosine	Similarity score
Summarization	BART	Short complaint summary

4.5 Data Analysis & Engineering Judgment

The performance of the system can be evaluated by analyzing the output of individual NLP modules.

For department and priority classification, accuracy, precision, recall, F1-score, and confusion matrices can be used.

For duplicate detection, similarity scores should be evaluated using known duplicate and non-duplicate complaint pairs.

Translation quality can be evaluated by checking whether the translated text preserves the original complaint meaning.

For summarization, generated summaries should be examined for clarity, relevance, and preservation of the original complaint issue.

Any incorrectly classified complaints should be analyzed to identify ambiguous language, insufficient training examples, translation errors, or closely related department categories.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Detect complaint language	FastText output	Fully Achieved
Translate complaint	NLLB output	Fully Achieved
Classify department	BERT prediction	Fully Achieved
Determine priority	Random Forest prediction	Fully Achieved
Detect duplicate complaint	Similarity score	Fully Achieved
Generate summary	BART output	Fully Achieved
Route complaint	Department dashboard	Fully Achieved

4.7 Strengths & Limitations

Strengths

- Automated complaint processing.
- Multilingual complaint support.
- Context-aware department classification.
- Automatic priority detection.
- Semantic duplicate detection.
- Automatic complaint summarization.
- Reduced manual workload.
- Centralized complaint tracking.
- Modular system architecture.

Limitations

- Performance depends on the quality and size of the training dataset.
- Large Transformer models require higher computational resources.
- Translation quality may differ between languages.
- Similar complaints from nearby but different locations may occasionally be identified as duplicates.
- New complaint categories may require model retraining.
- Human verification may still be necessary for highly critical complaints.

4.8 Project Planning, Roles & Budget

The project can be divided into the following development stages:

1. Problem identification and requirement analysis.
2. Dataset collection and preprocessing.
3. Language detection implementation.
4. Translation module implementation.
5. BERT department-classification model development.
6. Random Forest priority model development.
7. SBERT duplicate-detection implementation.
8. BART summarization implementation.
9. Frontend development.
10. Backend and database integration.
11. Testing and evaluation.
12. Documentation and final demonstration.

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The AI-Driven Citizen Support and Complaint Management Using Natural Language Processing project provides an intelligent approach for improving conventional citizen grievance-management systems.

The proposed system integrates multiple NLP and machine-learning algorithms to automate important complaint-processing activities. FastText is used for language identification, NLLB for multilingual translation, BERT for department classification, Random Forest for priority detection, SBERT with Cosine Similarity for duplicate detection, and BART for complaint summarization.

By combining these modules, the system can understand citizen complaints, categorize them, identify their level of urgency, detect repeated complaints, generate concise summaries, and forward complaints to the appropriate departments.

The proposed architecture reduces manual processing effort and provides a structured approach for handling large numbers of citizen complaints. The project also demonstrates how modern Natural Language Processing techniques can be integrated into a practical public-service application.

5.2 Future Work

The project can be extended by:

- Supporting additional Indian regional languages.
- Allowing citizens to submit voice complaints.
- Applying speech-to-text before NLP processing.
- Supporting image-based complaint analysis.
- Adding automatic location extraction from complaint text.
- Integrating GIS/maps for complaint visualization.
- Developing mobile applications.
- Providing automatic notifications through SMS or email.
- Increasing the size of the training dataset.
- Optimizing models for faster processing.
- Connecting the system with real government grievance portals.
- Developing analytics to identify frequently affected areas and recurring civic problems.

5.3 Professional Learning & Individual Reflection

The project provides learning in both academic and professional areas. From an academic perspective, it connects Natural Language Processing concepts such as language detection, text classification, semantic similarity, priority detection, and text summarization with a complete application. From a technical perspective, it introduces BERT for department classification, FastText and NLLB for multilingual processing, Random Forest for priority detection, SBERT for duplicate complaint detection, and BART for text summarization.

The development process also highlights the importance of understanding the complete complaint management workflow. A feature such as multilingual complaint processing is not only a translation task; it affects language detection, translation, classification, priority identification, duplicate checking, summarization, and final routing. This understanding improves the ability to design NLP systems as an integrated process rather than as a collection of independent models.

Student

Individual Reflection

Key Learning

Y Sai Kumar	Worked on department classification using BERT.	Learned contextual NLP classification.
I. Leela Sravan Kumar	Worked on multilingual complaint processing.	Learned language detection and translation.
T. Kiran Kumar	Worked on complaint priority detection using Random Forest.	Learned urgency and severity classification.
A. Mohith Krishna	Worked on duplicate complaint detection using SBERT.	Learned semantic similarity and comparison.
K. Varshan	Worked on complaint summarization using BART.	Learned text summarization and system integration.

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APPENDICES

App.tsx:

```
import React, { useState, useEffect } from 'react';
import { Header } from './components/Header';
import { Navigation, TabType } from './components/Navigation';
import { HomePage } from './pages/HomePage';
import { SubmitComplaintPage } from './pages/SubmitComplaintPage';
import { MyComplaintsPage } from './pages/MyComplaintsPage';
import { TrackComplaintPage } from './pages/TrackComplaintPage';
import { AdminDashboard } from './pages/AdminDashboard';
import { DepartmentDashboard } from './pages/DepartmentDashboard';
import { AcademicNLPExplorer } from './pages/AcademicNLPExplorer';
import { api } from './services/api';

export const App: React.FC = () => {
  const [activeTab, setActiveTab] = useState<TabType>('home');
  const [aiMode, setAiMode] = useState<string>('REAL');

  useEffect(() => {
    // Fetch AI system mode on load
    api.getSystemStatus()
      .then(res => setAiMode(res.ai_mode))
      .catch(() => setAiMode('REAL'));
  }, []);

  return (
    <div className="min-h-screen bg-slate-950 text-slate-100 flex flex-col font-sans">

      {/* Top Banner Header */}
      <Header aiMode={aiMode} />

      {/* Main Navigation Bar */}
      <Navigation activeTab={activeTab} setActiveTab={setActiveTab} />

      {/* Main Content Body */}
      <main className="flex-1 max-w-7xl w-full mx-auto px-4 sm:px-6 lg:px-8 pt-6">
        {activeTab === 'home' && <HomePage setActiveTab={setActiveTab} />}
        {activeTab === 'submit' && (
          <SubmitComplaintPage
            setActiveTab={setActiveTab}
          />
        )}
      </main>
    </div>
  );
}
```

```

    })
    {activeTab === 'my-complaints' && <MyComplaintsPage />}
    {activeTab === 'track' && <TrackComplaintPage />}
    {activeTab === 'admin' && <AdminDashboard />}
    {activeTab === 'department' && <DepartmentDashboard />}
    {activeTab === 'nlp-explorer' && <AcademicNLPExplorer />}
  </main>

  { /* Footer */ }
  <footer className="bg-slate-900/80 border-t border-slate-800/80 py-6 text-center text-xs
text-slate-400">
    <div className="max-w-7xl mx-auto px-4 flex flex-col sm:flex-row items-center
justify-between gap-2">
      <p>© 2026 AI-Driven Citizen Support & Complaint Management System.</p>
      <p className="font-mono text-slate-400">Natural Language Processing Academic
Capstone</p>
    </div>
  </footer>

</div>
);
};

export default App;

```